MSHI-GEO

Soil Microbial Respiration: A Three-Tier Monitoring Stack

Published EAB sensor work
MSHI-Geo manuscript queued for ESSD

[Author Name] · [School / Institution] · Genius Olympiad 2026

github.com/Sparkxt-0318/MSHI

01 · PROBLEM

Soil respiration has no high-resolution validator.

Soils release ~91 Pg C yr^{-1} to the atmosphere — second only to ocean exchange. Continental Rs upscaling models (Hashimoto 2015; Warner 2019; Stell 2021) report uncertainty as Monte-Carlo CIs on the global sum, not as held-out site-level transfer.

We characterise the cross-continental transfer gap directly: train an XGBoost on Asian Rs sites, test on a held-out US subset, report transfer R^2 + CI, and decompose the gap into SHAP drivers per region. The result clarifies what the standard feature stack can and cannot do at this scale, and where ground-truth biosensor measurement is needed.

02 · METHODS

Open data, gradient-boosted trees, held-out continent.

Training
  615 Asia (605 SRDB v5 + 10 COSORE)
  274 US (SRDB v5 + 10 COSORE)
  Target: log Rs_annual (g C m^{-2} yr^{-1})

Features (20)
  8 WorldClim 2.1 bioclim variables
  8 SoilGrids 2.0 5-15 cm topsoil layers
  4 engineered (C/N, clay/sand, pH-opt, aridity)

Model & validation
  XGBoost depth=3, n_est=250, regularised
  5-fold spatial-block CV (5° blocks), Asia
  Asia → US transfer R^2 + 2,000-iter bootstrap

03 · FRAMEWORK

One biology, three scales of measurement.

cm — **Electrochemical biosensor**

Electron-transfer current ∝ substrate oxidation. Centimetre footprint, continuous, low cost.

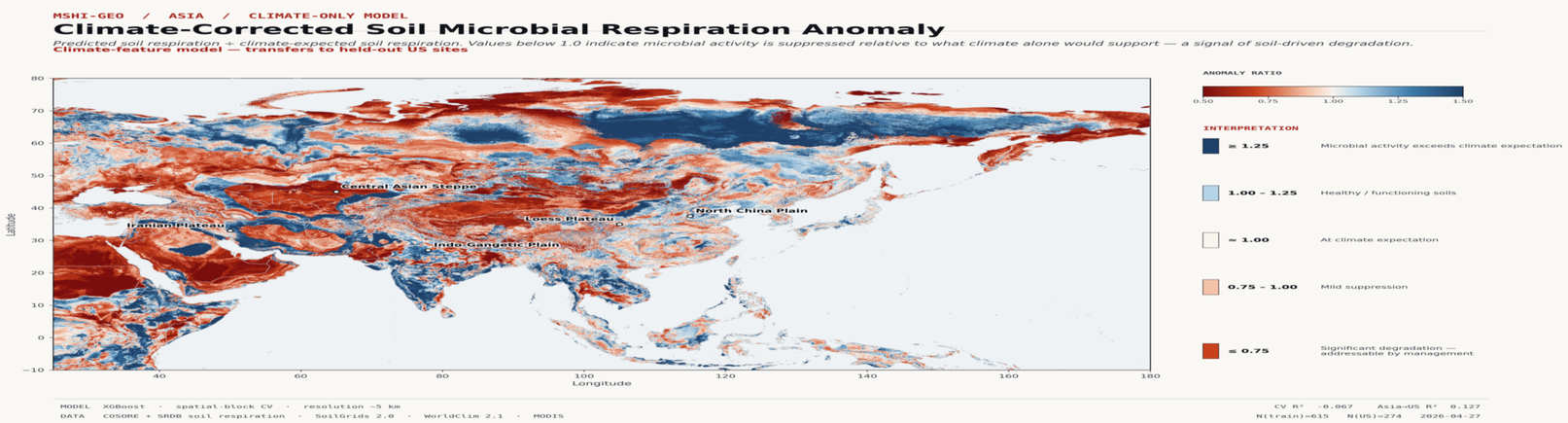
m — **Chamber + eddy-covariance**

Direct CO_2 flux measurement. Metre footprint, sparse, high instrument cost.

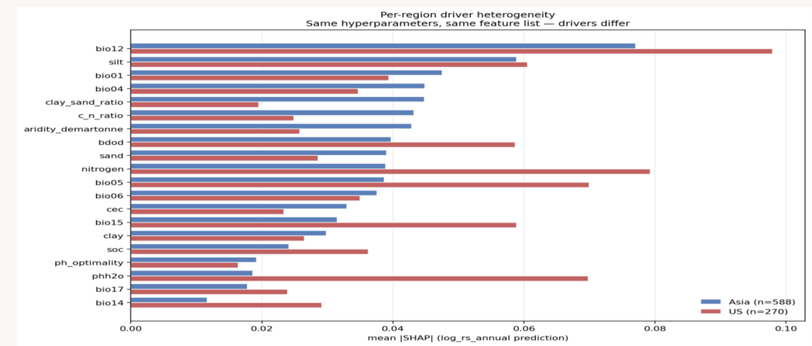
km — **Satellite + ML upscaling (this work)**

5 km grid. Climate + soil features. Transfer-tested.

04 · HERO MAP — climate-only model on Asia (R^2 = +0.127, CI excludes 0)

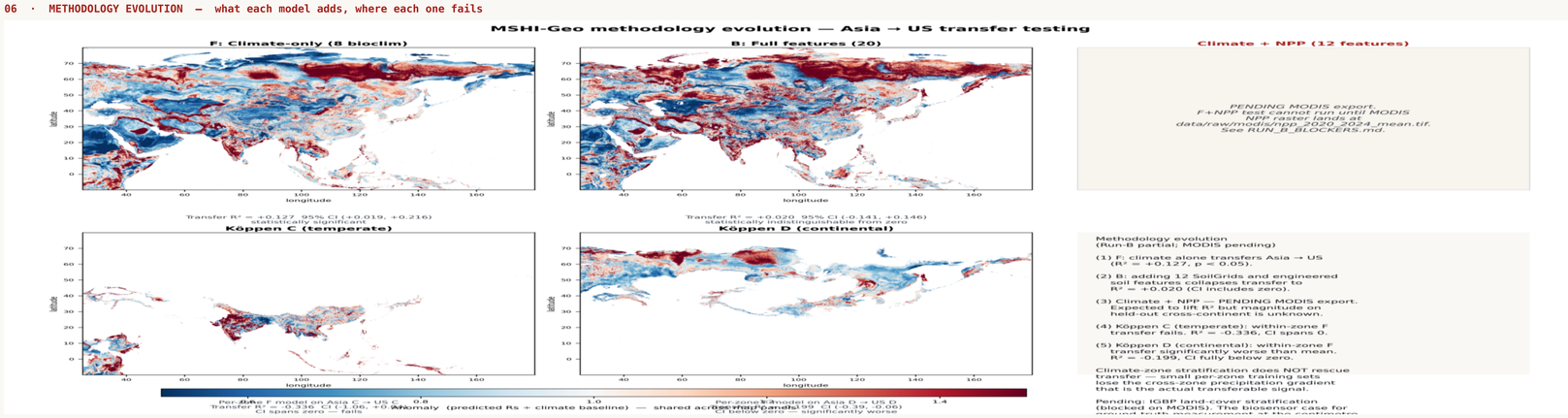


05 · REGIONAL DRIVER HETEROGENEITY

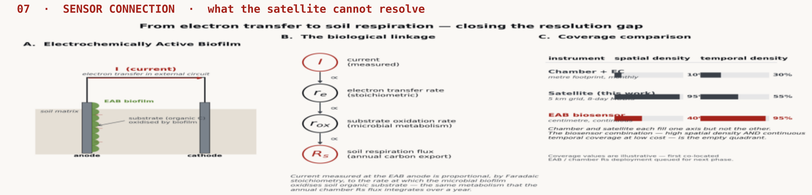


Same XGBoost trained separately on Asia and US. bio12 (annual precip) is the only feature with stable rank in both regions. clay/sand-ratio drops out of US top 8; nitrogen j

06 · METHODOLOGY EVOLUTION — what each model adds, where each one fails



07 · SENSOR CONNECTION · what the satellite cannot resolve



08 · KÖPPEN STRATIFICATION · doesn't help
Köppen C R^2 = -0.336 CI (-1.06, +0.03) spans 0
Köppen D R^2 = -0.199 CI (-0.39, -0.06) significantly negative
Cross-zone reference R^2 = +0.122 CI (+0.02, +0.22)

The cross-zone +0.127 IS the cross-zone precipitation gradient.
Stratifying it away leaves no transferable signal.

09 · ROADMAP · three deliverables, three months

1.	MODIS NPP integration	≤ 2 weeks
2.	ESSD manuscript submission	≤ 6 weeks
3.	Co-located biosensor pilot at SRDB site	≤ 12 weeks

REFERENCES

Bond-Lamberty 2010, 2020 · Fick & Hijmans 2017 · Hashimoto 2015 (10.5194/bg-12-4121-2015) · Jian 2021 (10.5194/essd-13-255-2021) · Poggio 2021 · Stell 2021 (10.1111/gcb.15666) · Warner 2019 (10.1029/2019GB006264) · Yao 2021 (10.1029/2020GB006918).